

Navigation resilience programme: Earth, Moon and Mars

Mission dossier: requirements, runs and verification matrix

MODELLED RESULTS - every number in this dossier is simulation output of the open Kshana engine, labelled **VALIDATED**, **MODELLED** or **PARTNER** as the engine labels it. This is an engineering study, not a certification and not acceptance by any authority.

Reference KP-MD-0001
Issue 1
Classification Unclassified
Issued 2026-09-30T16:13:45Z
Engine open Kshana 0.29.0, 75 scenario kinds
Overlay Kshana Pro 0.2.0, development build
Dossier SHA-256 49df1e0e78e07cc53649ff7d2ded07a53c02da1af508fce46660cb5da599eac9

12

Requirements

7

Met

4

Not met

1

Not verified

15

Open items

5 scenario runs and one trade study. 11 verdicts, one per requirement and linked run: 0 on a **VALIDATED** basis, 7 on a **MODELLED** basis, 4 on a **PARTNER** basis.



Verdict summary

12 requirements: 7 met, 4 not met, 1 not verified. Met on a MODELLED or PARTNER basis, and flagged: 7. With no run traced: 1. Open items to close before sign-off: 15.

Met: the criterion holds on every linked run. Not met: it fails on at least one. Not verified: a field is missing, a unit does not match, an array is not reduced, the run's files fail an integrity check, or no run is linked.

VALIDATED, MODELLED and PARTNER are the open engine's own labels for the capability behind a number; a verdict carries the weakest label it rests on.

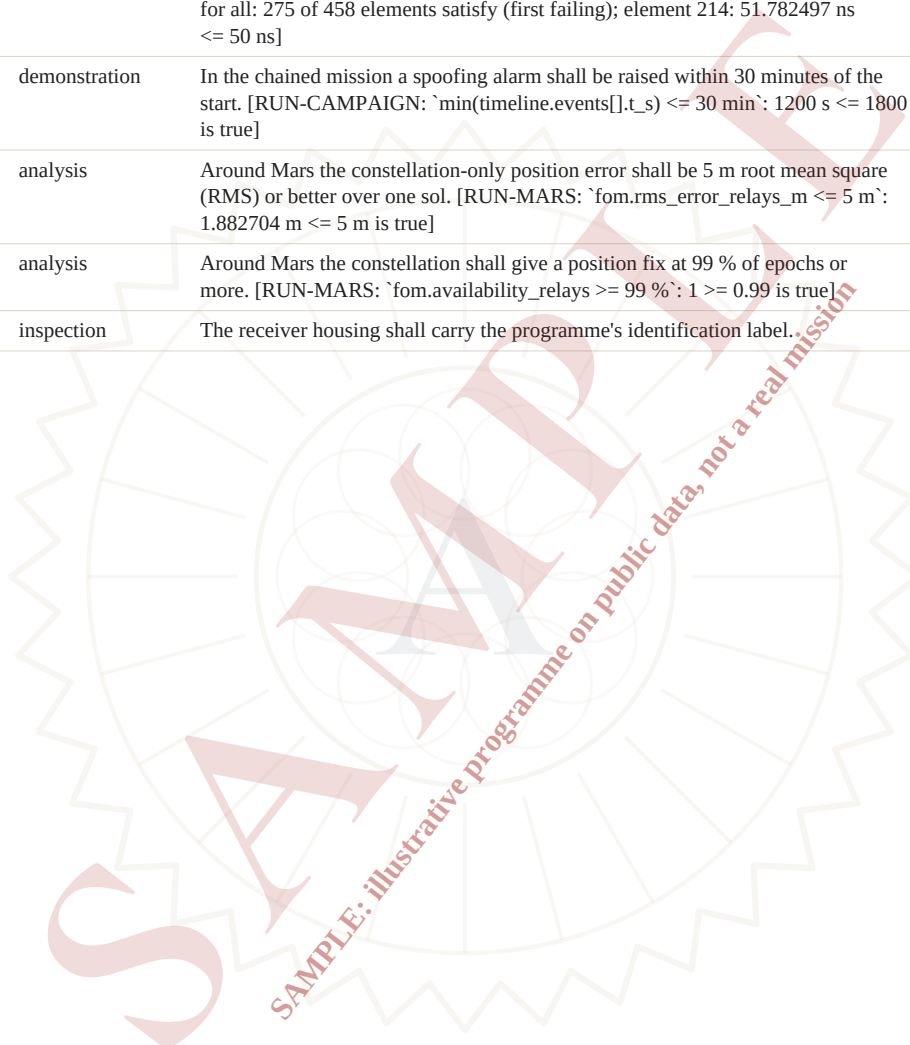
Verification matrix

Requirement	Criterion	Run	Verdict	Basis	Value read
LEO-001	ppp.cases[1].median_convergence_min <= 10 min	RUN-LEO	met	MODELLED	7.5 min
LEO-002	ppp.cases[0].median_convergence_min <= 10 min	RUN-LEO	not met	MODELLED	11.5 min
LEO-003	fusion.fused.rms_error_3d_m <= 0.5 m	RUN-LEO	met	MODELLED	0.434888 m
LUN-001	global.availability_pct >= 95 %	RUN-LUNAR	not met	MODELLED	21.3286 %
LUN-002	global.pdop.median <= 6	RUN-LUNAR	met	MODELLED	5.22
SPC-001	timeline.bands[3].min_cn0_dbhz >= 25 dB-Hz	RUN-SPECTRUM	met	PARTNER	44.0752 dB-Hz
SPC-002	timeline.bands[0].min_cn0_dbhz >= 25 dB-Hz	RUN-SPECTRUM	not met	PARTNER	3.23286 dB-Hz
CMP-001	timeline.channels.time_error_ns.values <= timeline.channels.guard_ns.values for all values	RUN-CAMPAIGN	not met	PARTNER	51.7825 ns
CMP-002	min(timeline.events[.t_s]) <= 30 min	RUN-CAMPAIGN	met	PARTNER	1200 s
MARS-001	fom.rms_error_relays_m <= 5 m	RUN-MARS	met	MODELLED	1.8827 m
MARS-002	fom.availability_relays >= 99 %	RUN-MARS	met	MODELLED	1
OPS-001	none	-	not verified	-	no run traced; method inspection

Requirements and what each verdict read

Requirement	Method	Text and evidence
LEO-001	analysis	With 240 low Earth orbit (LEO) satellites added, precise point positioning (PPP) shall converge within 10 minutes (median over the seeded runs). [RUN-LEO: `ppp.cases[1].median_convergence_min <= 10 min`: 7.5 min <= 10 min is true]
LEO-002	analysis	With GPS and Galileo alone, precise point positioning (PPP) shall converge within 10 minutes (median over the seeded runs). [RUN-LEO: `ppp.cases[0].median_convergence_min <= 10 min`: 11.5 min <= 10 min is false]
LEO-003	test	With the LEO layer the fused three-dimensional position error shall be 0.5 m root mean square (RMS) or better. [RUN-LEO: `fusion.fused.rms_error_3d_m <= 0.5 m`: 0.434888 m <= 0.5 m is true]
LUN-001	analysis	The lunar relay constellation shall give a position fix with a position dilution of precision (PDOP) of 6 or better over at least 95 % of the lunar surface and of the day. [RUN-LUNAR: `global.availability_pct >= 95 %`: 21.3286 % >= 95 % is false]

Requirement	Method	Text and evidence
LUN-002	analysis	Where the lunar relay constellation gives a fix, the median position dilution of precision (PDOP) shall be 6 or better. [RUN-LUNAR: `global.pdop.median <= 6`: 5.22 <= 6 is true]
SPC-001	analysis	Under the scripted L-band interference, the GPS L5 carrier-to-noise density ratio (C/N0) shall stay at or above 25 dB-Hz. [RUN-SPECTRUM: `timeline.bands[3].min_cn0_dbhz >= 25 dB-Hz`: 44.075187 dB-Hz >= 25 dB-Hz is true]
SPC-002	analysis	Under the scripted L-band interference, the GPS L1 coarse/acquisition (C/A) carrier-to-noise density ratio (C/N0) shall stay at or above 25 dB-Hz. [RUN-SPECTRUM: `timeline.bands[0].min_cn0_dbhz >= 25 dB-Hz`: 3.232859 dB-Hz >= 25 dB-Hz is false]
CMP-001	analysis	In the chained jamming, spoofing and holdover mission the clock time error shall stay within its 50 ns guard at every timeline step. [RUN-CAMPAIGN: `timeline.channels.time_error_ns.values <= timeline.channels.guard_ns.values` for all: 275 of 458 elements satisfy (first failing); element 214: 51.782497 ns <= 50 ns]
CMP-002	demonstration	In the chained mission a spoofing alarm shall be raised within 30 minutes of the start. [RUN-CAMPAIGN: `min(timeline.events[.t_s] <= 30 min`: 1200 s <= 1800 s is true]
MARS-001	analysis	Around Mars the constellation-only position error shall be 5 m root mean square (RMS) or better over one sol. [RUN-MARS: `fom.rms_error_relays_m <= 5 m`: 1.882704 m <= 5 m is true]
MARS-002	analysis	Around Mars the constellation shall give a position fix at 99 % of epochs or more. [RUN-MARS: `fom.availability_relays >= 99 %`: 1 >= 0.99 is true]
OPS-001	inspection	The receiver housing shall carry the programme's identification label.





Run RUN-LEO

LEO-PNT end to end, generic 1080 km constellation

Low Earth orbit (LEO) positioning, navigation and timing chain on a generic 1080 km constellation: signal, pass, navigation message, fused fix, precise point positioning.

Kind leo-pnt-chain
Stamp seed 5, engine 0.29.0
Engine summary LEO-PNT end to end, generic 1080 km constellation signal generic-l (https://www.itu.int/pub/R-REG-RR), tracked pilot at 10.230 Mchip/s, C/N0 offset -2.57 dB pass LEO-1 1080 km: tracked C/N0 peak 58.2 dB-Hz, code jitter median 0.050 m; C/N0 line 35.6 + 24.3 sin(el) dB-Hz message kepler-rac 300 s fit / 300 s update: representation SISRE 0.907 mm RMS (orbit 0.005 mm), with 0.250 m orbit determination: 0.250 m handed on fused fix: GNSS only PDOP 1.47 RMS 3D 1.60 m -> GNSS + LEO PDOP 1.06 RMS 3D 0.43 m PPP convergence: GNSS only 11.5 min, 240 LEO 7.5 min 11 values handed between stages. MODELLED.
Scenario hash not stamped by this kind
Result SHA-256 0be8376208d92b6d431b0aeb26146b0fff7a2b4e3f1ecdb18b37e99a16573a6d

Run label: MODELLED - the chain hands each stage's output to the next; the stages keep their own labels (leo-signal, leo-pass, leo-navmsg, leo-pvt and leo-ppp rows of the verification matrix).

Requirements verified on this run

Requirement	Verdict	Basis
LEO-001	met	MODELLED
LEO-002	not met	MODELLED
LEO-003	met	MODELLED

Headline figures: scalars the result marks as computed

Field	Value	Unit	What it is
fusion.gnss.median_pdop	1.47453	1	median position dilution of precision, GNSS only
fusion.fused.median_pdop	1.06478	1	median position dilution of precision, GNSS + LEO
fusion.fused.availability	1	1	fraction of epochs with a fused fix
fusion.fraction_epochs_with_leo	1	1	fraction of epochs with at least one LEO satellite
pass.peak_cn0_dbhz	60.7342	dB-Hz	total C/N0 at the pass peak
pass.peak_tracked_cn0_dbhz	58.1678	dB-Hz	tracked-component C/N0 at the pass peak
pass.duration_above_mask_s	845	s	time the chain satellite is above the pass mask
pass.max_elevation_deg	75	deg	maximum elevation of the chain satellite's pass
ppp.cases[0].fraction_converged	1	1	GNSS only: fraction of runs that converged
ppp.cases[1].fraction_converged	1	1	240 LEO: fraction of runs that converged

Outputs the result marks with another provenance class

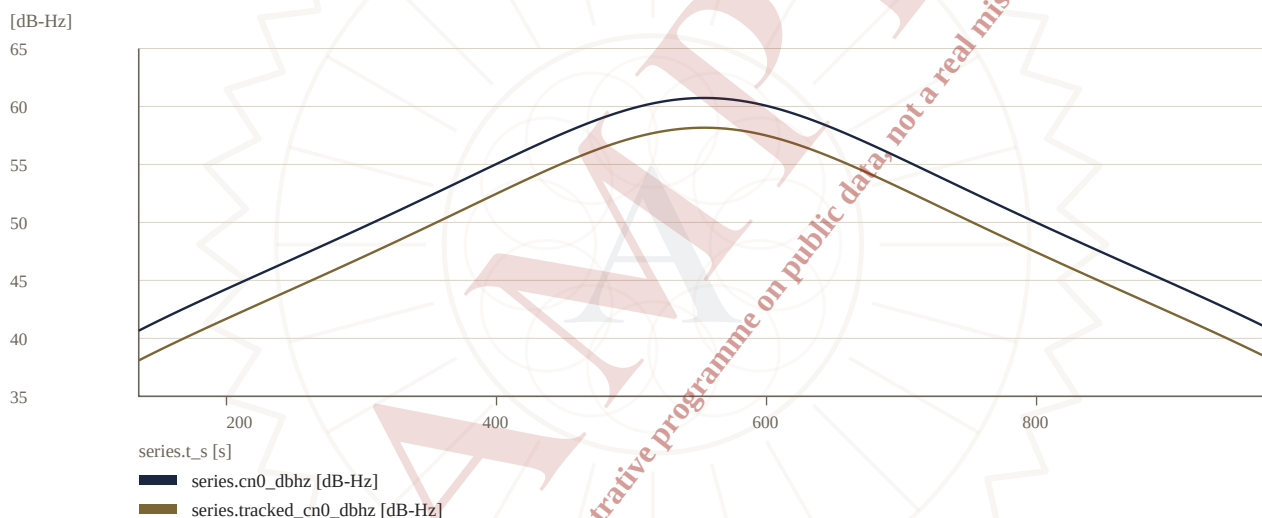
Field	Value	Unit	Provenance and what it is
fusion.fused.median_sigma_3d_m	0.433791	m	modelled: median formal 3D one-sigma, GNSS + LEO
fusion.fused.rms_error_3d_m	0.434888	m	modelled: root-mean-square 3D error, GNSS + LEO
fusion.gnss.median_sigma_3d_m	1.57565	m	modelled: median formal 3D one-sigma, GNSS only

Field	Value	Unit	Provenance and what it is
fusion.gnss.rms_error_3d_m	1.60216	m	modelled: root-mean-square 3D error, GNSS only
fusion.leo.median_sigma_3d_m	0.450815	m	modelled: median formal 3D one-sigma, LEO only
fusion.leo.rms_error_3d_m	0.484131	m	modelled: root-mean-square 3D error, LEO only
fusion.systems[0].sigma_pr_30deg_m	1.25122	m	modelled: pseudorange one-sigma at 30 deg elevation
fusion.systems[1].sigma_pr_30deg_m	1.13822	m	modelled: pseudorange one-sigma at 30 deg elevation
fusion.systems[2].sigma_pr_30deg_m	0.257124	m	modelled: pseudorange one-sigma at 30 deg elevation
navmsg.representation_sisre_rms_m	0.000906565	m	modelled: the message's representation error against its truth orbit and clock, root mean square
navmsg.sisre_max_m	0.00242187	m	modelled: largest signal-in-space range error, orbit and clock
navmsg.sisre_orbit_rms_m	5.26208e-6	m	modelled: orbit-only signal-in-space range error, root mean square

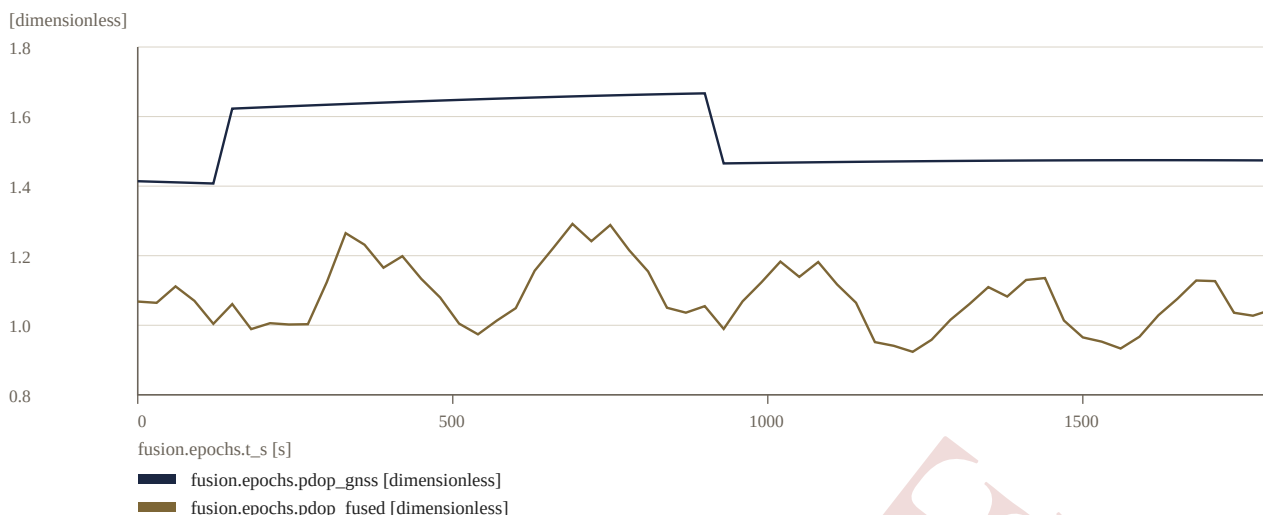
6 more are listed in dossier.json.

In dossier.json for this run: 42 computed scalars, 8 computed columns, 18 outputs of another provenance class, 13 echoed inputs and constants (listed as inputs, never as figures), 0 numeric fields with no units entry.

Carrier-to-noise density over the pass



Position dilution of precision, GNSS alone and with the LEO layer



Capability labels, from the run's report.json

Capability	Label	Role
Reproducibility & software assurance	MODELLED	exercised
Band-limited closed forms for any ranging signal: power in band and early-late code-tracking jitter against published values, with the Gabor bandwidth and offset spectral separation cross-checked	VALIDATED	exercised
Low Earth orbit positioning, navigation and timing signal designs: code tracking, acquisition, GNSS compatibility and a band trade for any system	MODELLED	exercised
Tropospheric amplitude scintillation on an Earth-space link	VALIDATED	exercised
First-order ionospheric delay per band, the ionosphere-free combination and free-space loss	VALIDATED	exercised
Maximum Doppler a static user sees from a LEO or MEO orbit	VALIDATED	exercised
A LEO-PNT pass and its per-band link budget against the MEO GNSS satellites in view	MODELLED	exercised
Named LEO-PNT system presets with stated sources, and a system-agnostic engine	MODELLED	exercised
Global-average signal-in-space range error weights for any orbit altitude	VALIDATED	exercised
Galileo ICD broadcast-ephemeris user algorithm as the base of a LEO navigation message	VALIDATED	exercised
LEO broadcast-ephemeris fitter and signal-in-space range error versus fit interval and update period	MODELLED	exercised
Joint GNSS and LEO pseudorange positioning with inter-system biases and per-signal error models	MODELLED	exercised
Precise point positioning convergence with GNSS only and with LEO augmentation	MODELLED	exercised
One LEO-PNT system end to end: signal design, pass link budget, navigation message and fused positioning, each stage's output handed to the next	MODELLED	exercised

Not modelled, from the run's report.json

- The positioning stage models C/N0 as a straight line in the sine of the elevation between the mask and the zenith; the chain fits that line to the pass's tracked C/N0, so a pass-specific shape (an isoflux plateau, a beam edge) is reduced to two numbers. [result `limitations[0]`]
- The SISRE handed to the positioning stages is the navigation message's representation error against a Kshana-integrated truth orbit, plus the scenario's stated od_sisre_m in root-sum-square; the message stage itself models no orbit determination or prediction. [result `limitations[1]`]
- The positioning stage computes its own delay-lock-loop jitter from the handed-on C/N0 with an unlimited-bandwidth formula; the pass reports the band-limited jitter of the actual design beside it. [result `limitations[2]`]
- The precise point positioning stage takes the message's SISRE as the LEO orbit-and-clock error; its code and phase noise stay the stage's own inputs. [result `limitations[3]`]
- One satellite, the first whose band carries the design, stands for the whole LEO system in the pass and message



stages. [result `limitations[4]`]

- MODELLED - the chain hands each stage's output to the next; the stages keep their own labels (leo-signal, leo-pass, leo-navmsg, leo-pvt and leo-ppp rows of the verification matrix). [result `label`]

10 more are listed in dossier.json and in the run's report.





Run RUN-LUNAR

lunar-relay-constellation (constellation-design)

A 14-satellite lunar relay and navigation constellation, scored as a coverage and dilution-of-precision map over the whole Moon for one day.

Kind	constellation-design
Stamp	seed not recorded, engine 0.29.0
Engine summary	constellation-design: 14 satellites (Lunar relay) around the Moon; availability 21.33% (PDOP <= 6 at 10 deg mask), worst site 0.00%, median PDOP 5.22, mean 3.2 in view; 648 grid points x 48 epochs (MODELLED)
Scenario hash	not stamped by this kind
Result SHA-256	52b51b8113951d901f1ead50bf49a3806478cfa59461c3ebde899c5fe6801a09

Run label: MODELLED - two-body Keplerian orbits (optional secular J2) around a spherical body, geometric visibility above an elevation mask, all-in-view dilution of precision with one receiver clock per constellation unless clock = "common"; no signal power, health or terrain. The Walker geometry and the published presets are checked against the Galileo OS SDD, the GLONASS ICD and the GPS SPS Performance Standard slot tables, and the GPS baseline global DOP statistics against SPS PS Appendix B (see docs/VERIFICATION-MATRIX.md)

Requirements verified on this run

Requirement	Verdict	Basis
LUN-001	not met	MODELLED
LUN-002	met	MODELLED

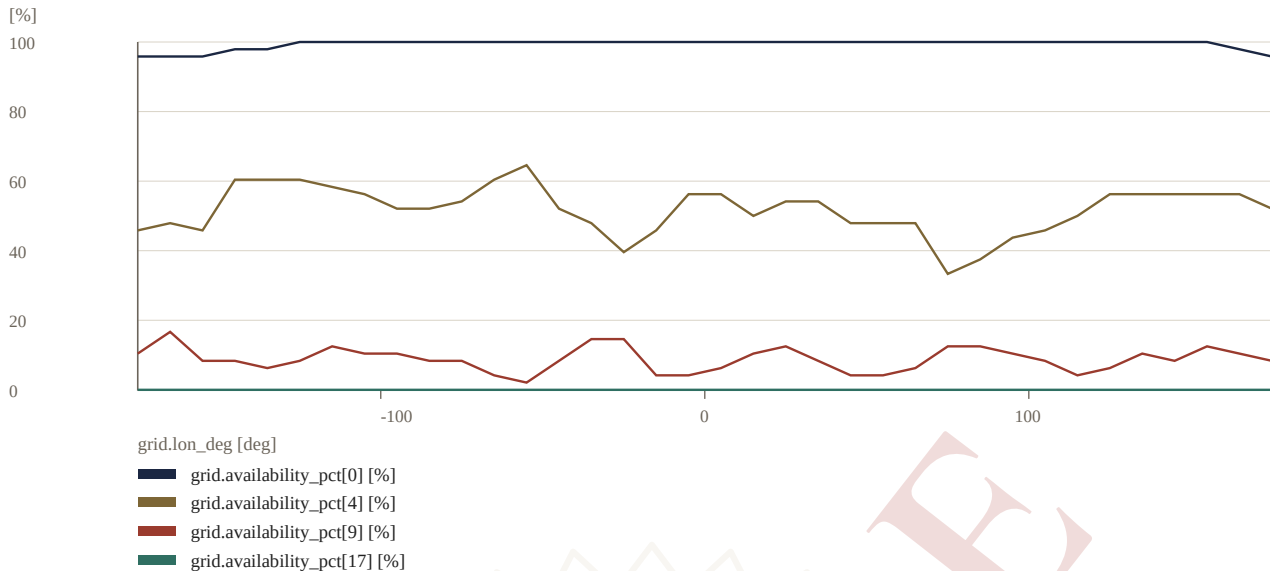
Headline figures: scalars the result marks as computed

Field	Value	Unit	What it is
global.availability_pct	21.3286	%	area-weighted share of time-space points with a fix and PDOP at or below pdop_threshold
global.worst_site_availability_pct	0	%	availability of the grid cell with the lowest availability
global.fix_pct	38.1248	%	area-weighted share of time-space points with enough satellites in view for a non-singular fix
global.pdop.median	5.22		area-weighted median of the position dilution of precision (PDOP) over the time-space points with a fix (quantiles resolved to 0.005)
global.pdop.p95	41.2176	1	area-weighted 95th percentile of the position dilution of precision (PDOP) over the time-space points with a fix (quantiles resolved to 0.005)
global.mean_visible	3.1858	count	area-weighted mean number of satellites above the mask
global.min_visible	0	count	fewest satellites above the mask at any time-space point
total_satellites	14	count	satellites across every constellation of the run

In dossier.json for this run: 44 computed scalars, 16 computed columns, 0 outputs of another provenance class, 12 echoed inputs and constants (listed as inputs, never as figures), 0 numeric fields with no units entry.



Availability along four latitudes (85 S, 45 S, 5 N, 85 N), by longitude



Capability labels, from the run's report.json

Capability	Label	Role
Reproducibility & software assurance	MODELLED	exercised
Walker constellation geometry and the published nominal slots of GPS, Galileo and GLONASS	VALIDATED	exercised
Global dilution of precision of the GPS baseline constellation	VALIDATED	exercised
Coverage and dilution-of-precision maps for arbitrary multi-constellation designs at scale, around any central body	MODELLED	exercised

Not modelled, from the run's report.json

- MODELLED - two-body Keplerian orbits (optional secular J2) around a spherical body, geometric visibility above an elevation mask, all-in-view dilution of precision with one receiver clock per constellation unless clock = "common"; no signal power, health or terrain. The Walker geometry and the published presets are checked against the Galileo OS SDD, the GLONASS ICD and the GPS SPS Performance Standard slot tables, and the GPS baseline global DOP statistics against SPS PS Appendix B (see docs/VERIFICATION-MATRIX.md) [result 'label']
- MODELLED: two-body orbits with optional secular J2, a spherical body, geometric visibility only (no signal power, health or terrain), and the relative phase between different systems, which is not a snapshot of any real date. [kind catalogue ('kshana kinds --json')]
- What the run answers: where on the Moon four or more of these satellites are in view above a 10 deg mask, the PDOP/HDOP/VDOP/GDOP per 10 deg cell, and the share of the day with PDOP at or below 6. MODELLED: two-body orbits around a spherical Moon (no Earth third-body perturbation, which a real frozen-orbit design depends on), geometry only. [scenario file comment, lines 18 to 21]
- Reproducibility & software assurance is MODELLED, not validated: a sub-claim is externally checked, but the whole capability composes modelled pieces, so the capability stays Modelled. [verification matrix (docs/MODELLED-RATIONALE.md)]
- Coverage and dilution-of-precision maps for arbitrary multi-constellation designs at scale, around any central body is MODELLED, not validated: checked against its own closed-form / analytic identity - catches transcription and coefficient errors, but is not an external oracle. [verification matrix (docs/MODELLED-RATIONALE.md)]



Run RUN-SPECTRUM

I-band-waterfall-jamming (spectrum)

The L band under a scripted sequence of three jammers: which navigation signals each one takes away.

Kind	spectrum
Stamp	seed 7, engine 0.29.0
Engine summary	scenario spectrum 5 bands 3 jammers noise floor -202.0 dBW/Hz worst band gps-l1ca min C/N0 3.2 dB-Hz (J/S 60.2 dB)
Scenario hash	5b6d91fbe8900aef98e62df8e7b51704462854a7f3522535e7942c37d59d4184
Result SHA-256	75455b50f5df107fe353e13f154b464cad17b55ec605f1707dc9ab6a47b8e389

Run label: MODELLED: an L-band power spectral density built from closed-form signal spectra (VALIDATED against the textbook main-lobe widths and spectral separation coefficients), jammer spectra and a kT noise floor, reduced to J/S and effective C/N0 by the spectral separation coefficient. The jammer powers, timeline and front-end bandwidths are scenario inputs, not measurements.

Requirements verified on this run

Requirement	Verdict	Basis
SPC-001	met	PARTNER
SPC-002	not met	PARTNER

Headline figures: scalars the result marks as computed

Field	Value	Unit	What it is
timeline.bands[0].min_cn0_dbhz	3.23286	dB-Hz	gps-l1ca: lowest row C/N0
timeline.bands[1].min_cn0_dbhz	4.69394	dB-Hz	galileo-e1: lowest row C/N0
timeline.bands[2].min_cn0_dbhz	36.857	dB-Hz	gps-l2c: lowest row C/N0
timeline.bands[3].min_cn0_dbhz	44.0752	dB-Hz	gps-l5: lowest row C/N0
timeline.bands[4].min_cn0_dbhz	46.9752	dB-Hz	galileo-e5a: lowest row C/N0
timeline.bands[0].tracking_fraction	0.166667	1	gps-l1ca: fraction of rows at or above the tracking threshold
timeline.bands[1].tracking_fraction	0.5	1	galileo-e1: fraction of rows at or above the tracking threshold
timeline.bands[2].tracking_fraction	1	1	gps-l2c: fraction of rows at or above the tracking threshold
timeline.bands[3].tracking_fraction	1	1	gps-l5: fraction of rows at or above the tracking threshold
timeline.bands[4].tracking_fraction	1	1	galileo-e5a: fraction of rows at or above the tracking threshold
timeline.bands[0].worst_js_db	60.2394	dB	gps-l1ca: largest row in-band J/S; null when no jammer reaches the band
timeline.bands[1].worst_js_db	67.1328	dB	galileo-e1: largest row in-band J/S; null when no jammer reaches the band

Outputs the result marks with another provenance class

Field	Value	Unit	Provenance and what it is
bands[0].nominal_cn0_dbhz	43.4752	dB-Hz	closed-form: un-jammed C/N0 of the tracked component: its power minus the noise density
bands[0].tracked_power_dbw	-158.5	dBW	closed-form: power the C/N0 and J/S are referred to: the signal power for a plain band; for a designed signal the tracked component's share of the unfiltered power, P_s / η

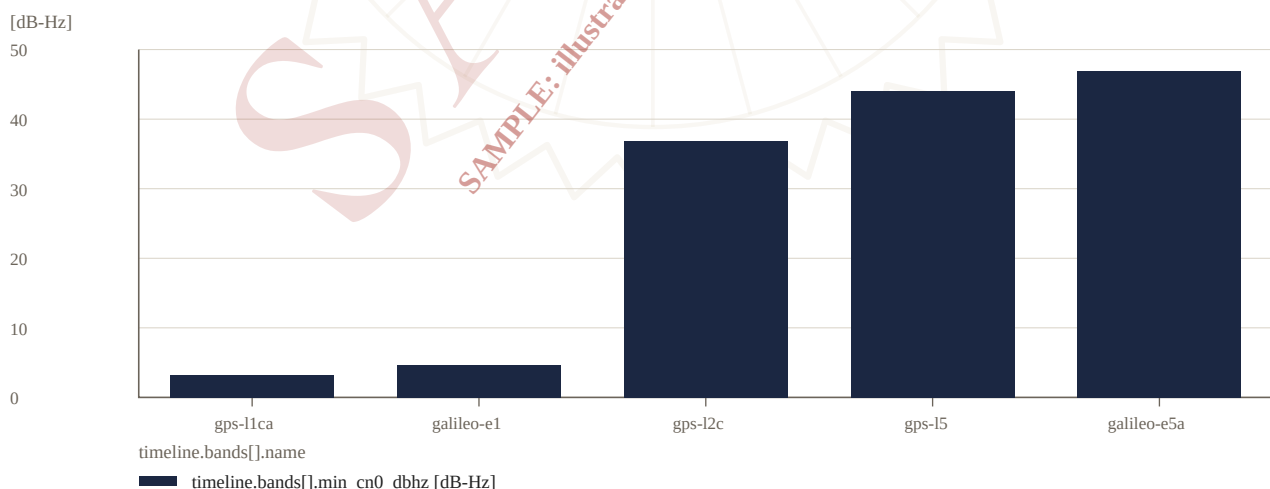


Field	Value	Unit	Provenance and what it is
bands[1].nominal_cn0_dbhz	44.9752	dB-Hz	closed-form: un-jammed C/N0 of the tracked component: its power minus the noise density
bands[1].tracked_power_dbw	-157	dBW	closed-form: power the C/N0 and J/S are referred to: the signal power for a plain band; for a designed signal the tracked component's share of the unfiltered power, P_s / η
bands[2].nominal_cn0_dbhz	41.9752	dB-Hz	closed-form: un-jammed C/N0 of the tracked component: its power minus the noise density
bands[2].tracked_power_dbw	-160	dBW	closed-form: power the C/N0 and J/S are referred to: the signal power for a plain band; for a designed signal the tracked component's share of the unfiltered power, P_s / η
bands[3].nominal_cn0_dbhz	44.0752	dB-Hz	closed-form: un-jammed C/N0 of the tracked component: its power minus the noise density
bands[3].tracked_power_dbw	-157.9	dBW	closed-form: power the C/N0 and J/S are referred to: the signal power for a plain band; for a designed signal the tracked component's share of the unfiltered power, P_s / η
bands[4].nominal_cn0_dbhz	46.9752	dB-Hz	closed-form: un-jammed C/N0 of the tracked component: its power minus the noise density
bands[4].tracked_power_dbw	-155	dBW	closed-form: power the C/N0 and J/S are referred to: the signal power for a plain band; for a designed signal the tracked component's share of the unfiltered power, P_s / η
iq.comparison.integrated_power_ratio	0.999883	1	internal-consistency: Welch total power over model total power across the band, tones included
iq.comparison.mean_abs_difference_db	5.43937	dB	internal-consistency: mean absolute Welch-minus-model difference over the compared bins

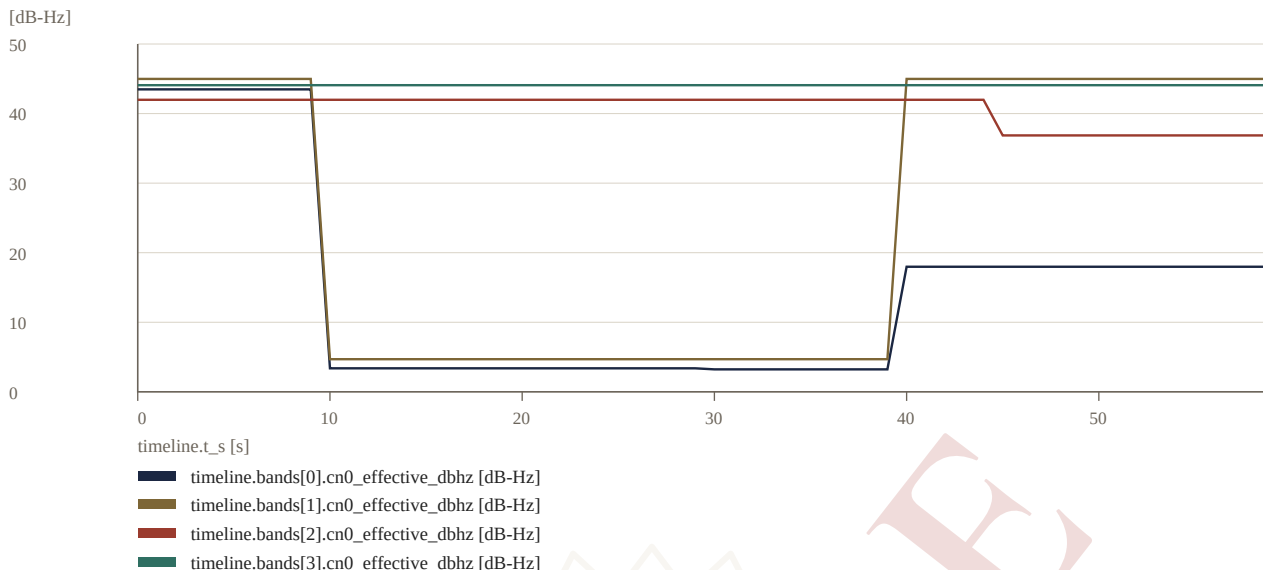
8 more are listed in dossier.json.

In dossier.json for this run: 67 computed scalars, 15 computed columns, 20 outputs of another provenance class, 55 echoed inputs and constants (listed as inputs, never as figures), 0 numeric fields with no units entry.

Lowest carrier-to-noise density of each band



Carrier-to-noise density over time: GPS L1 C/A, Galileo E1, GPS L2C, GPS L5



Capability labels, from the run's report.json

Capability	Label	Role
Reproducibility & software assurance	MODELLED	exercised
Navigation RF payload & antenna hardware design	PARTNER	relied on, not provided
Closed-form L-band signal power spectral densities and spectral separation coefficients	VALIDATED	exercised
L-band spectrum waterfall with per-band J/S and effective C/N0 under a scripted jammer timeline	MODELLED	exercised
SigMF recording input and output, and Welch spectral estimates of complex IQ	MODELLED	exercised
Multi-band spectrum waterfall (UHF, L, S, C) with designed signals and per-band jammers	MODELLED	exercised

Not modelled, from the run's report.json

- spreading-code line structure (for example the 1 kHz lines of C/A): tones are scored against the smooth envelope [result `not\_modelled[0]`]
 - automatic gain control, quantisation, pulse blanking and notch filtering [result `not\_modelled[1]`]
 - the receive-antenna pattern toward the jammer beyond one gain figure [result `not\_modelled[2]`]
 - intra-system multiple-access interference between satellites of one band [result `not\_modelled[3]`]
 - GLONASS G1/G2, BeiDou B1/B2 and Galileo E6 [result `not\_modelled[4]`]
 - a designed signal's emissions outside its transmit bandwidth: its spectrum is truncated there, so no out-of-band emission reaches a neighbouring band [result `not\_modelled[5]`]
- 9 more are listed in dossier.json and in the run's report.



Run RUN-CAMPAIGN

Jamming, spoofing, holdover and integrity: a chained mission

A chained mission on one timeline: nominal, jamming, spoofing, holdover, integrity alarm, recovery.

Kind	campaign
Stamp	seed 20260928, engine 0.29.0
Engine summary	campaign f87a4a0e0bab Jamming, spoofing, holdover and integrity: a chained mission chain: 6 phases over 4570 s, 2 events, alarm raised on 2770 s of the grid 18 member runs (MODELLED)
Scenario hash	f87a4a0e0babe2ea4b7e8adc7739cf564e270dc1c261d2ff48386766bf9d1ba1
Result SHA-256	cb8d821da6eabd70e638b5ad5eb5103efc6f69ab63c78f97d34e7d5d2cb5879a

Run label: MODELLED composition of existing scenario kinds. Every number is read from a real run of the named kind; the chaining (additive carry of a channel across a phase boundary, zero-order hold onto the timeline grid, a phase ended at a computed time) is a modelling choice, and each phase carries the label of the kind that produced it.

Requirements verified on this run

Requirement	Verdict	Basis
CMP-001	not met	PARTNER
CMP-002	met	PARTNER

Headline figures: scalars the result marks as computed

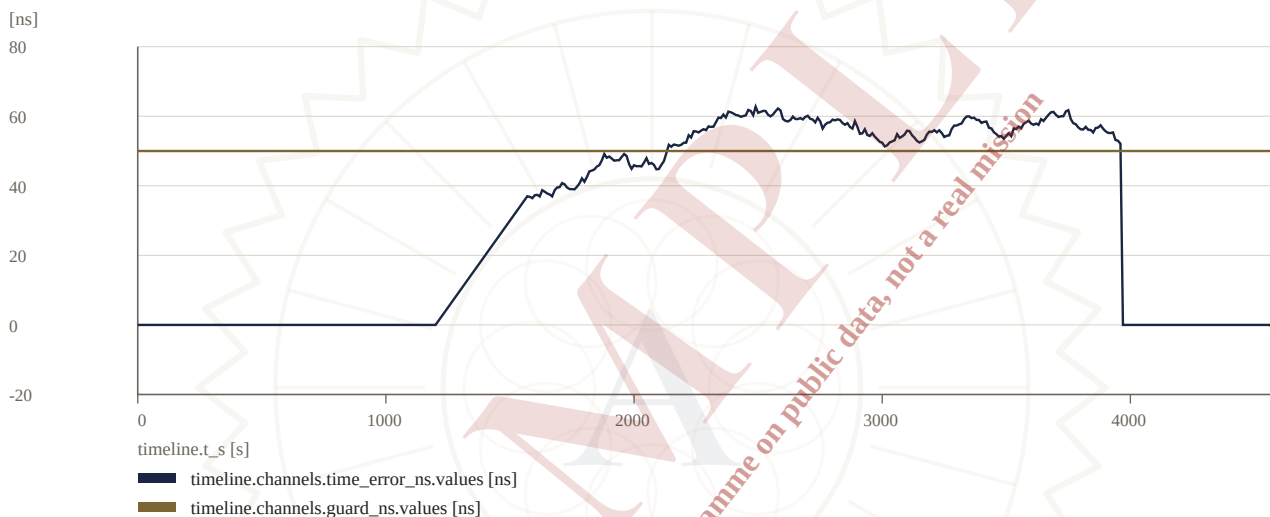
Field	Value	Unit	What it is
reproducibility.runs_total	18	count	member scenario runs the campaign dispatched
timeline.duration_s	4570	s	mission length: the sum of the phase lengths as run
timeline.events[0].t_s	1200	s	RF spoofing detector alarms (fused consistency, power and signal-quality monitors): mission time of the event, read from the run that raised it
timeline.events[1].t_s	1570	s	clock-aided spoofing monitor alarms: mission time of the event, read from the run that raised it
timeline.phases[0].t0_s	0	s	nominal: mission time at which the phase starts (the previous phase's end)
timeline.phases[0].t1_s	600	s	nominal: mission time at which the phase ends: t0_s plus duration_s, or earlier when end_at fired
timeline.phases[1].t0_s	600	s	jamming: mission time at which the phase starts (the previous phase's end)
timeline.phases[1].t1_s	1200	s	jamming: mission time at which the phase ends: t0_s plus duration_s, or earlier when end_at fired
timeline.phases[2].t0_s	1200	s	spoofing: mission time at which the phase starts (the previous phase's end)
timeline.phases[2].t1_s	1570	s	spoofing: mission time at which the phase ends: t0_s plus duration_s, or earlier when end_at fired
timeline.phases[3].carried.time_error_ns	37	ns	holdover: value of time_error_ns at the end of the previous phase, added to this phase's time_error_ns
timeline.phases[3].t0_s	1570	s	holdover: mission time at which the phase starts (the previous phase's end)



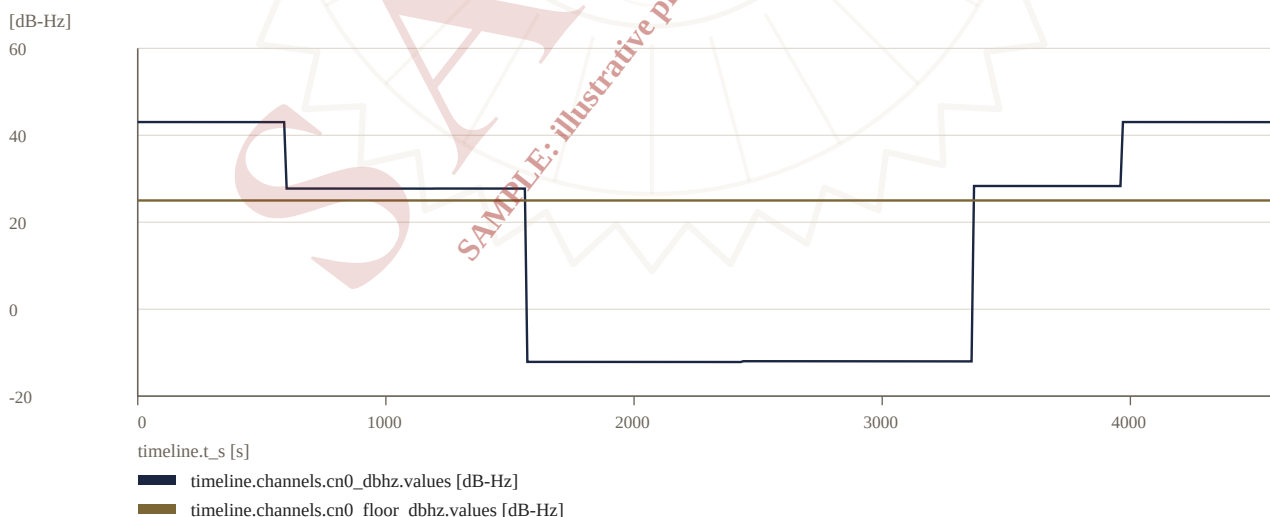
Field	Value	Unit	What it is
timeline.phases[3].t1_s	3370	s	holdover: mission time at which the phase ends: t0_s plus duration_s, or earlier when end_at fired
timeline.phases[4].carried.time_error_ns	59.5803	ns	integrity-alarm: value of time_error_ns at the end of the previous phase, added to this phase's time_error_ns
timeline.phases[4].t0_s	3370	s	integrity-alarm: mission time at which the phase starts (the previous phase's end)
timeline.phases[4].t1_s	3970	s	integrity-alarm: mission time at which the phase ends: t0_s plus duration_s, or earlier when end_at fired
timeline.phases[5].t0_s	3970	s	recovery: mission time at which the phase starts (the previous phase's end)
timeline.phases[5].t1_s	4570	s	recovery: mission time at which the phase ends: t0_s plus duration_s, or earlier when end_at fired

In dossier.json for this run: 18 computed scalars, 11 computed columns, 0 outputs of another provenance class, 2 echoed inputs and constants (listed as inputs, never as figures), 0 numeric fields with no units entry.

Clock time error against its guard



Carrier-to-noise density against the tracking floor



Capability labels, from the run's report.json



Capability	Label	Role
Frequency stability characterisation	VALIDATED	exercised
Integrity (RAIM/ARAIM/SBAS)	VALIDATED	exercised
GNSS geometry / dilution of precision (DOP)	VALIDATED	exercised
RAIM/ARAIM integrity statistical kernel (chi^2 / non-central chi^2 / normal laws)	VALIDATED	exercised
GNSS-denied clock holdover	MODELLED	exercised
Onboard clock state estimation	MODELLED	exercised
Quantum inertial sensor performance	MODELLED	exercised
GNSS/INS sensor fusion	MODELLED	exercised
GNSS-denied jamming resilience	MODELLED	exercised
Spoofing detection	MODELLED	exercised
Reproducibility & software assurance	MODELLED	exercised
Navigation RF payload & antenna hardware design	PARTNER	relied on, not provided
A chained mission across scenario kinds on one shared timeline	MODELLED	exercised

Not modelled, from the run's report.json

- MODELLED composition of existing scenario kinds. Every number is read from a real run of the named kind; the chaining (additive carry of a channel across a phase boundary, zero-order hold onto the timeline grid, a phase ended at a computed time) is a modelling choice, and each phase carries the label of the kind that produced it. [result `label`]
 - MODELLED: the additive carry and the zero-order hold are modelling choices, and each phase carries the label of the kind that ran it. [kind catalogue (`kshana kinds --json`)]
 - MODELLED: the chaining is a modelling choice (additive carry, zero-order hold), and each phase is as good as the kind that ran it. [scenario file comment, lines 30 to 31]
 - GNSS-denied clock holdover is MODELLED, not validated: checked against a separate implementation in this same codebase - independent of the unit under test, but not externally authoritative. [verification matrix (docs/MODELLED-RATIONALE.md)]
 - Onboard clock state estimation is MODELLED, not validated: checked against a separate implementation in this same codebase - independent of the unit under test, but not externally authoritative. [verification matrix (docs/MODELLED-RATIONALE.md)]
 - Quantum inertial sensor performance is MODELLED, not validated: a sub-claim is externally checked, but the whole capability composes modelled pieces, so the capability stays Modelled. [verification matrix (docs/MODELLED-RATIONALE.md)]
- 5 more are listed in dossier.json and in the run's report.

Run RUN-MARS

mars-orbit-pnt (body-pnt)

A 300 km Mars orbiter positioned by twelve navigation satellites and a two-way range from Earth, over one sol.

Kind	body-pnt
Stamp	seed 7, engine 0.29.0
Engine summary	Positioning around Mars - orbiter user, 12 relays at 10000 km, 296 epochs Earth light time 338.3 s one-way, 676.6 s round trip; Sun-Earth-body angle 175.4 deg availability: 1.000 constellation only, 1.000 with the Earth range (Earth in view 0.811) median PDOP 1.375 ; RMS error 1.883 m constellation only, 1.629 m with the Earth range median formal sigma 1.375 m constellation only, 1.304 m with the Earth range
Scenario hash	not stamped by this kind
Result SHA-256	99ae45fc9aa3c074767da83f01bd4d078b48cefb4e3119195191e19c05b90842

Run label: MODELLED: two-body relay orbits with J2 secular drift, geometric visibility, Gaussian range noise and an instantaneous measurement model around a body from the published constants; not validated against a mission's navigation data. The Earth-to-body geometry inherits the solar-system ephemeris labels.

Requirements verified on this run

Requirement	Verdict	Basis
MARS-001	met	MODELLED
MARS-002	met	MODELLED

Headline figures: scalars the result marks as computed

Field	Value	Unit	What it is
fom.rms_error_relays_m	1.8827	m	root-mean-square position error of the seeded constellation-only fixes
fom.rms_error_with_earth_m	1.62856	m	root-mean-square position error of the seeded fixes with the Earth range
fom.availability_relays	1	1	fraction of epochs with a converged constellation-only fix (at least four relays)
fom.availability_with_earth	1	1	fraction of epochs with a converged fix once the Earth range is added
fom.earth_visibility	0.810811	1	fraction of epochs the Earth is above the body's limb for the user
fom.median_pdop	1.37453	1	median constellation-only position dilution of precision over epochs with four or more relays
fom.mean_relays_visible	6.78378	count	mean number of relays in view
earth_to_body.one_way_light_time_s	338.316	s	Newtonian light time from the body (retarded) to the Earth at the epoch
earth_to_body.sun_separation_deg	175.439	deg	Sun-Earth-body angle at the Earth; small near a solar conjunction, where plasma and Shapiro delay grow

Outputs the result marks with another provenance class

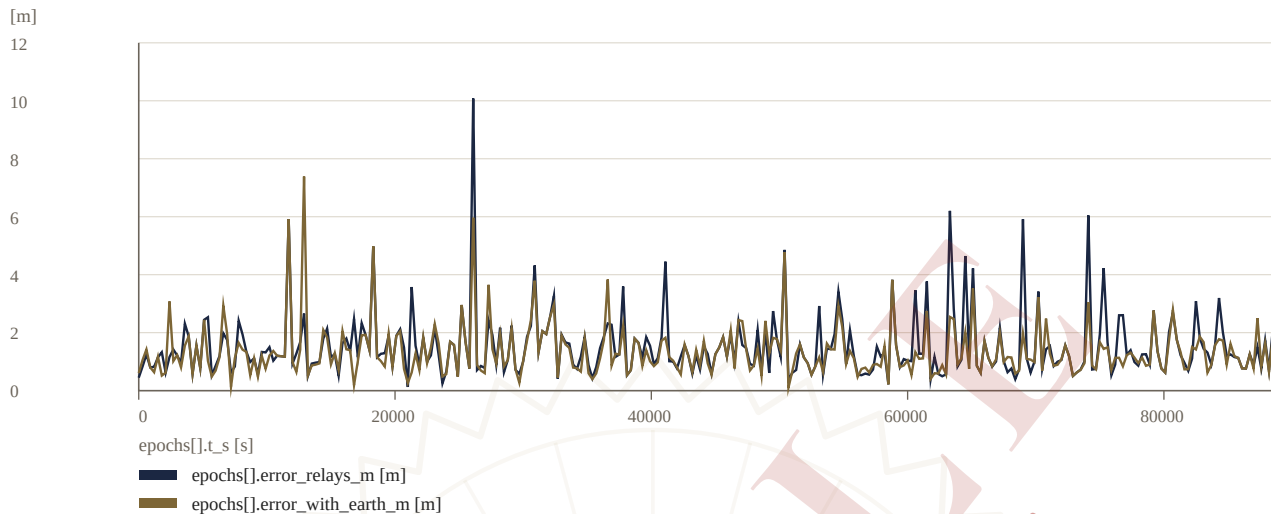
Field	Value	Unit	Provenance and what it is
body.sidereal_rotation_period_h	24.623	h	derived: 360 degrees over the IAU prime-meridian rate; negative for retrograde rotation
earth_to_body.shapiro_delay_s	5.14806e-6	s	closed-form: the Sun's one-way Shapiro delay on the Earth-to-body path



Field	Value	Unit	Provenance and what it is
relay_period_s	47039.3	s	closed-form: two-body period of the relay orbit, $2 \pi \sqrt{a^3/GM}$

In dossier.json for this run: 18 computed scalars, 10 computed columns, 3 outputs of another provenance class, 9 echoed inputs and constants (listed as inputs, never as figures), 0 numeric fields with no units entry.

Position error of the seeded fix, with and without the Earth range



Capability labels, from the run's report.json

Capability	Label	Role
Reproducibility & software assurance	MODELLED	exercised
Positioning around any solar-system body with a local constellation and a deep-space link from Earth	MODELLED	exercised

Not modelled, from the run's report.json

- MODELLED: two-body relay orbits with J2 secular drift, geometric visibility, Gaussian range noise and an instantaneous measurement model around a body from the published constants; not validated against a mission's navigation data. The Earth-to-body geometry inherits the solar-system ephemeris labels. [result `label`]
- MODELLED: the relay orbits, the noise levels and the instantaneous measurement model are modelling choices, not validated against a mission's navigation data; the body constants are the published values cited in body.rs, and the Earth geometry inherits the solar-system ephemeris labels. [kind catalogue Pkshana kinds --json`]
- MODELLED. The relay orbits, the 1 m range noise and the instantaneous measurement model are modelling choices; nothing here is validated against a mission's navigation data. The Mars constants are the published values cited in src/body.rs. [scenario file comment, lines 20 to 22]
- Reproducibility & software assurance is MODELLED, not validated: a sub-claim is externally checked, but the whole capability composes modelled pieces, so the capability stays Modelled. [verification matrix (docs/MODELLED-RATIONALE.md)]
- Positioning around any solar-system body with a local constellation and a deep-space link from Earth is MODELLED, not validated: checked against its own closed-form / analytic identity - catches transcription and coefficient errors, but is not an external oracle. [verification matrix (docs/MODELLED-RATIONALE.md)]

Trade study

Lunar relay: south-biased 14 satellites against north-and-south 22

Metric global.availability_pct (higher is better), read from each variant's own run.

Rank	Variant	Metric value	Scenario hash
2	south-biased-14	21.3286	not stamped by this kind
1	north-and-south-22	60.629	not stamped by this kind

Recommended architecture: 'north-and-south-22' - best 'global.availability_pct' (higher is better) across 2 comparable variant(s).

The trade's own assurance gate: Fail. 1 of 1 assurance checks failed: multi-criteria decision present.

The trade study is a Kshana Pro kind. Each variant is an open-engine scenario run unchanged; the trade itself is not reproduced by the free engine.





Open items: 15

Category	Requirement	Run	What must be closed
MODELLED basis	LEO-001	RUN-LEO	met on a MODELLED basis: the capability behind this figure is not externally validated in the open verification matrix
not met	LEO-002	RUN-LEO	`ppp.cases[0].median_convergence_min <= 10 min`: 11.5 min <= 10 min is false
method	LEO-003	-	the declared method is test; simulation evidence counts as analysis (ECSS-E-ST-10-02C), so the test itself still needs its own record
MODELLED basis	LEO-003	RUN-LEO	met on a MODELLED basis: the capability behind this figure is not externally validated in the open verification matrix
not met	LUN-001	RUN-LUNAR	`global.availability_pct >= 95 %`: 21.3286 % >= 95 % is false
MODELLED basis	LUN-002	RUN-LUNAR	met on a MODELLED basis: the capability behind this figure is not externally validated in the open verification matrix
PARTNER basis	SPC-001	RUN-SPECTRUM	met on a PARTNER basis: the capability behind this figure is not externally validated in the open verification matrix
not met	SPC-002	RUN-SPECTRUM	`timeline.bands[0].min_cn0_dbhz >= 25 dB-Hz`: 3.232859 dB-Hz >= 25 dB-Hz is false
not met	CMP-001	RUN-CAMPAIGN	`timeline.channels.time_error_ns.values <= timeline.channels.guard_ns.values` for all: 275 of 458 elements satisfy (first failing); element 214; 51.782497 ns <= 50 ns
method	CMP-002	-	the declared method is demonstration; simulation evidence counts as analysis (ECSS-E-ST-10-02C), so the demonstration itself still needs its own record
PARTNER basis	CMP-002	RUN-CAMPAIGN	met on a PARTNER basis: the capability behind this figure is not externally validated in the open verification matrix
MODELLED basis	MARS-001	RUN-MARS	met on a MODELLED basis: the capability behind this figure is not externally validated in the open verification matrix
MODELLED basis	MARS-002	RUN-MARS	met on a MODELLED basis: the capability behind this figure is not externally validated in the open verification matrix
no evidence	OPS-001	-	no run is traced to this requirement
manual verification	OPS-001	-	no machine-checkable criterion: verify by inspection and record the result in the sign-off table

Reproducibility record

The engine pins floating-point results per platform and per build: another operating system or processor architecture, or an unoptimised build of the engine, can give slightly different floating-point values in some fields. Then the hash differs, and a field-by-field comparison of the two result files shows by how much. `kshana-pro mission check <dir> --rerun` repeats every run on the embedded engine and compares the hashes. This dossier was produced by a development build of Kshana Pro, which compiles the embedded engine at the release optimisation level; a dossier for delivery is produced by a release build.

RUN-LEO

Command kshana leo-pnt-end-to-end.toml
Run it in a copy of runs/RUN-LEO
Scenario SHA-256 4a0286b53161bfff2257bb693b77f73c20632df077f0eb84a459d5a32ef21ebce
Result SHA-256 0be8376208d92b6d431b0aeb26146b0fff7a2b4e3f1ecdb18b37e99a16573a6d
Report SHA-256 491fbe75c2102c3ab7e743d15380208aa02e0784b46442cabaa3ea1002a30995

The free open Kshana engine repeats this run: in a directory that holds only a copy of leo-pnt-end-to-end.toml, the command writes leo-pnt-end-to-end.result.json, and with a release build of engine 0.29.0 on macos/aarch64 that file has the SHA-256 recorded here.



RUN-LUNAR

Command kshana lunar-relay-constellation.toml
Run it in a copy of runs/RUN-LUNAR
Scenario SHA-256 d626dcb56d0749b58bae19d446a5d869cecc3126ce499d0862c2be2a2665dff3
Result SHA-256 52b51b8113951d901f1ead50bf49a3806478cfa59461c3ebde899c5fe6801a09
Report SHA-256 7e9189e745f2af6e7522f7410a4051ee7e4f9f3127e6c322ff7c5d44f7a8981c

The free open Kshana engine repeats this run: in a directory that holds only a copy of lunar-relay-constellation.toml, the command writes lunar-relay-constellation.result.json, and with a release build of engine 0.29.0 on macos/aarch64 that file has the SHA-256 recorded here.

RUN-SPECTRUM

Command kshana l-band-waterfall-jamming.toml
Run it in a copy of runs/RUN-SPECTRUM
Scenario SHA-256 204830758faac62c51e3b396ff425442fee7d781378790c9c4ca7173f6c4973d
Result SHA-256 75455b50f5df107fe353e13f154b464cad17b55ec605f1707dc9ab6a47b8e389
Report SHA-256 c0e95e61afe712a6fc2d16a2462ec9a1b27e4821256eb3b2b1774a766c642573

The free open Kshana engine repeats this run: in a directory that holds only a copy of l-band-waterfall-jamming.toml, the command writes l-band-waterfall-jamming.result.json, and with a release build of engine 0.29.0 on macos/aarch64 that file has the SHA-256 recorded here.

RUN-CAMPAIGN

Command kshana campaign-jam-spoof-holdover-integrity.toml
Run it in a copy of runs/RUN-CAMPAIGN
Scenario SHA-256 cec1e10f8eb478bae58acebe9c2f06cfccf6572f6f6895dc633fb409b8d8ee5f
Result SHA-256 cb8d821da6eabd70e638b5ad5eb5103efcfff69ab63c78f97d34e7d5d2cb5879a
Report SHA-256 78b6967c3a50e336ea2f0734ed23fc2337117504db8eec37886b5e7591c3f23f

The free open Kshana engine repeats this run: in a directory that holds only a copy of campaign-jam-spoof-holdover-integrity.toml, the command writes campaign-jam-spoof-holdover-integrity.result.json, and with a release build of engine 0.29.0 on macos/aarch64 that file has the SHA-256 recorded here.

RUN-MARS

Command kshana mars-orbit-pnt.toml
Run it in a copy of runs/RUN-MARS
Scenario SHA-256 813ff12e5f8119636bd43bf164c08f061b97b6bb5e5da9362bec98f1e1d2f8e8
Result SHA-256 99ae45fc9aa3c074767da83f01bd4d078b48cefb4e3119195191e19c05b90842
Report SHA-256 7edad03bd8937da9716329a9b389801a8c7ba232369cda7ad3e99c13be54341f

The free open Kshana engine repeats this run: in a directory that holds only a copy of mars-orbit-pnt.toml, the command writes mars-orbit-pnt.result.json, and with a release build of engine 0.29.0 on macos/aarch64 that file has the SHA-256 recorded here.

TRADE

Command kshana-pro run trade-lunar-relay.toml --license <token>
Run it in a copy of runs/TRADE
Scenario SHA-256 03b3f722d635808127a5927740e47d3470113a94fabf28007132bcadca02888e
Result SHA-256 a645049be5743f74f489920239964c0ae025e50ee532d263f3611c3ae6c3154d
Report SHA-256 no report

'pro-trade-study' is a Kshana Pro kind: the free engine does not run it. The command repeats it under a licence that carries the kind, and 'kshana-pro mission check <dir> --rerun' compares the result's SHA-256. The open engine writes no report for a Pro kind, so this run's integrity rests on the result hash alone.



Inputs

File	SHA-256
programme.toml	b035b22ca5e56470388f11f40a6590704671a3fba43388f9b58921d9275f6339
requirements.csv	b0866379aa6f5e8680ca0026cf9bc42167325b3e2336f9b8edf33980e843d619
leo-pnt-end-to-end.toml	4a0286b53161bfff2257bb693b77f73c20632df077f0eb84a459d5a32ef21ebce
lunar-relay-constellation.toml	d626dcb56d0749b58bae19d446a5d869cecc3126ce499d0862c2be2a2665dff3
l-band-waterfall-jamming.toml	204830758faac62c51e3b396ff425442fee7d781378790c9c4ca7173f6c4973d
campaign-jam-spoof-holdover-integrity.toml	cec1e10f8eb478bae58acebe9c2f06cfcf6572f6f6895dc633fb409b8d8ee5f
mars-orbit-pnt.toml	813fff12e5f8119636bd43bf164c08f061b97b6bb5e5da9362bec98f1e1d2f8e8
trade-lunar-relay.toml	03b3f722d635808127a5927740e47d3470113a94fabf28007132bcadca02888e

Sign-off

Role	Name	Signature	Date
Prepared by			
Checked by			
Approved by			

Limitations

- Simulation evidence counts as analysis (ECSS-E-ST-10-02C, the European Cooperation for Space Standardization verification standard). For a requirement whose method is test, inspection or demonstration, a met criterion supports the requirement but does not replace the test, inspection or demonstration; it stays an open item.
- A label grades the capability as the open engine's verification matrix records it, not this programme's scenario inputs: a VALIDATED capability does not make an illustrative input measured.
- A figure is a scalar the result's own units block marks `computed`. A numeric field with no units entry is counted and not shown.
- Charts are drawn only for the paths the programme names. The chart the engine drew for each run is kept beside its result.
- A scenario is run from its own text. A data file a scenario names by relative path is not copied into the dossier.
- The PDF is identical between two builds except for the trailer identifier, which the PDF library fills with random characters.
- The dossier hash, the manifest and the ledger are not signed: they show that the files agree with each other, not who issued them. Pass the dossier hash to the recipient by a separate route when origin matters.
- Nothing in this dossier is a certification, a qualification or an acceptance by any authority.

Abbreviations: ECSS - European Cooperation for Space Standardization; HTML - HyperText Markup Language; JSON - JavaScript Object Notation; PDF - Portable Document Format; PNT - positioning, navigation and timing; SHA-256 - Secure Hash Algorithm, 256-bit; UTC - Coordinated Universal Time, the time scale of the issue stamp.